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Genetic Variation and Molecular Typing of *Xanthomonas oryzae* pv. *oryzae* Strains in Malaysian Rice Granaries

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Bacterial blight (BB), caused by *Xanthomonas oryzae* pv. *oryzae* (Xoo), poses a major threat to rice production in Malaysian rice granaries. Recent outbreaks between 2013 and 2017 have highlighted the severity of the disease in various rice-growing regions. The dynamic genetic structure of Xoo, combined with varying environmental conditions, presents a challenge in accurately characterizing the virulence diversity of the pathogen. Therefore, our study aimed to investigate the genetic diversity of Xoo strains to develop effective breeding strategies for BB resistance. We collected 100 strains of Xoo from eight ricegrowing regions in Malaysia and employed Repetitive Extragenic Palindromic PCR (REP-PCR) using REP, ERIC, and BOX primers, as well as an IS1112-based PCR approach using JEL primers, to analyze their genetic diversity. The ERIC primers generated the highest number of distinct PCR bands, followed by JEL, REP, and BOX primers, with band sizes ranging from 420bp to 7Kbp. Cluster analysis using combined Rep-PCR data revealed the presence of four distinct lineages of Xoo strains, with 12 haplotypes identified at 80% similarity index. Lineage 4 was widely distributed, consisting of 72 Xoo strains, with haplotype 11 being prevalent in BB hotspot areas. The IS1112-based dendrogram identified four lineages with 11 haplotypes, and lineage 3 exhibited wide distribution, comprising 69 strains. Notably, the grouping of haplotypes was influenced by the sampling time and geographical locations of the strains. The Rep-PCR approach was found to be more precise and effective in distinguishing Xoo strains from different geographic regions, particularly in BB hotspot areas. Interestingly, Xoo strains from the northern regions of Malaysia, which are recognized as significant disease-prone areas, exhibited a high degree of genetic diversity. These findings provide valuable insights for the management of Xoo population and the development of disease-resistant rice varieties. Understanding the genetic diversity of Xoo strains enables researchers to design targeted breeding strategies to enhance BB resistance.